

Use of Verbal Descriptors, Thermal Scores and Electrical Pulp Testing as Predictors of Tooth Pain Before and After Application of Benzocaine Gels Into Cavities of Teeth with Pulpitis

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A double-blind pilot study was conducted on 27 consenting human volunteers who had irreversible pulpitis associated with persistent toothache pain from open carious lesions. Formulations tested contained either 0, 10%, or 20% benzocaine and were identified only by a numbered code. Before the experiment started, a small amount of a known 5% benzocaine gel was placed for 1 minute on the tongue of each patient to assure a sensation of numbness within the oral cavity. Then the test tooth was washed with a gentle stream of warm water and dried with gauze. A randomly selected test medication was placed into the open cavity and around the gingival margins for 5 minutes. Pre- and posttreatment tests were conducted at the following timed intervals: 0, 5, 15, 30, 45, 60, 75 and 90 minutes. The tests included degree of pain (rated: 0 = none, 1 = mild, 2 = moderate, 3 = severe); electrical pulp testing (EPT) by a modified, voltage-ramping instrument; and ice water testing (0.5 mL directed quickly onto sound enamel of the tooth and rated: 0 to 4, with 4 being intolerable). After testing, or when pain returned to baseline, endodontic procedures were performed. There was a significant increase ($p < 0.032$, Fisher exact test) in subjects obtaining pain relief, rated by verbal descriptors, from the benzocaine gels

(14 out of 18 improved) compared to placebo (3 out of 9 improved). It was concluded that: 1) benzocaine gels are effective formulations for temporary relief of toothache pain, 2) there were no statistical differences in EPT scores between teeth having pulpitis and control teeth, 3) there were no correlations between direction of EPT scores and pain relief, 4) cold water testing was a good predictor of whether or not a tooth had pulpitis, and 5) changes in cold water testing scores after treatment could not be correlated to relief of pain according to verbal descriptors. The effectiveness of benzocaine in relieving toothache pain verifies previous studies; however, a difference between 10% and 20% benzocaine could not be demonstrated probably because of two factors: 1) the present experiment had a small sample size, and 2) there was no direct measurement of duration of local anesthesia.

The anesthetic benzocaine was introduced by Ritsert in 1890 as an alternative to cocaine.^{1,2} It has been widely used as a topical anesthetic since 1903.³⁻⁵ It is a p-aminobenzoic acid ester that lacks a cationic amino terminus.¹ This makes benzocaine poorly soluble in water, and a poor candidate for parenteral use.¹ Benzocaine is absorbed from ointments through the mucosa and into the blood stream, where it is rapidly destroyed by plasma cholinesterase.² These properties make benzocaine low in toxic potential and especially useful for topical anesthesia

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within the oral cavity.¹⁻⁶ Although allergic reactions to benzocaine have been reported following topical application, such reports are rare.⁶ When used as a topical anesthetic, benzocaine is usually incorporated into ointments or into a water-soluble base such as polyethylene glycol or propylene glycol. The topical anesthesia obtained from preparations of benzocaine has a rapid onset of action, but a short duration. If benzocaine remains in contact with tissues, the anesthetic effect may be more prolonged.⁷

Benzocaine has been sold over-the-counter (OTC) since 1926 for the temporary relief of toothache. The FDA Panel of OTC Dentifrices and Dental Care Agents⁷ recognized that benzocaine was available in a variety of preparations for professional or OTC use in concentrations from 6% to 22%,⁷ but its use as a toothache relief agent was classified in Category III (safe but efficacy studies required). Subsequently, Sveen et al. in 1982⁸ documented effectiveness in relieving toothache pain. They reported that 7.5% benzocaine in propylene glycol gel was an effective agent against persistent toothache pain that resulted from dental caries. Other investigators⁹ have supported benzocaine in the same formulation for relief of toothache pain.

The present pilot study was a double-blind controlled clinical trial testing two concentrations of benzocaine gels (10% and 20%) and placebo for relief of the toothache pain of irreversible pulpitis associated with an open carious lesion or with a lost filling. All teeth in the study were scheduled for vital pulpectomy. Toothache pain and relief were judged by verbal descriptor scoring and an attempt was made to correlate relief with changes in EPT and thermal cold scores.

METHODS AND MATERIALS

The study was double-blind using identical tubes of gel medications provided by the manufacturer (Orajel, Del Commerce, Farmingdale, NY). Tubes were randomly coded with four digit numbers; the preparations were indistinguishable in appearance, taste, viscosity, etc. An equal number of placebo, 10% benzocaine, and 20% benzocaine gels were provided. The total number of tubes was 27, 9 of each preparation. The order of assigning the medication was by a random draw with neither the investigator nor the patient aware of the agent being used. The code was maintained by the company and made available to the investigators at the completion of the study.

Patients selected for the study were judged to have pain from irreversible pulpitis due to an open cavity. The following patients were not included: those below 17 years old, those above 65 years old, those in poor health, those

with periapical or periodontal abscess, and those having used analgesics within 6 hours prior to entry.

Twenty-seven subjects volunteered for the experiment and signed a consent form. The group consisted of 19 females and 8 males; 14 patients were black and 13 were white. The age range was 17 to 41 years with a median age of 28. The treatment groups were rather evenly divided in composition by age, sex, and race so that no biases could have been introduced by demographic factors. The investigators were instructed to choose an anatomically similar control tooth that had the least possible restorative work: (in most cases there were no restorations in the controls). Tests were performed at baseline on experimental and control teeth. The tests consisted of: 1) verbal descriptors of pain converted to digital scores (severe = 3, moderate = 2, mild = 1, none = 0), 2), triplicate EPT scores determined with an Analytic Technology pulp tester (Analytic Technology, Redmond WA), modified so that the subject could discontinue the current at threshold, and 3) 0.5 mL of ice water, directed quickly onto an intact enamel area of the isolated tooth (care was taken to avoid squirting the ice water into the cavity), and rated by patient response on the following scale: 4 = intolerable discomfort, 3 = severe, 2 = moderate, 1 = mild, and 0 = none (the four point scale for cold testing was found to be useful for testing teeth with sensitive dentin in other studies).¹⁰

The experimental tooth selection included 14 molars, 12 premolars, and 1 canine. Controls contained 21 teeth matched with the same anatomical type, three premolars paired with canines, one canine paired with a premolar and two molars paired with premolars. Analysis of variance (ANOVA) testing of EPT scores by tooth type and by tooth pairings indicated no introduction of bias due to these factors; molars, premolars, and canines showed no statistically significant difference in EPT at baseline in this study.

To assure that all patients felt numbness whether receiving an active medication or a placebo, a small amount of 5% benzocaine gel on a saturated swab was placed against the tongue on the same side as the offending tooth. After 1 minute, the area was wiped with a dry gauze and the offending tooth cavity was gently rinsed with warm water. A tube of test medication was then randomly selected and its number recorded. The test medication was applied into the tooth cavity and onto the gingival tissues surrounding the tooth and allowed to remain in the cavity for 5 minutes. The patient was requested to indicate when the pain subsided, and the time of onset was recorded. The three discomfort tests were repeated after 5, 15, 30, 45, 60, 75 and 90 minutes. The experiment usually lasted 90 minutes, but could be terminated anytime after 15 minutes if no pain relief was obtained or if the pain returned to baseline level or higher.

A tooth was considered to have pain relief if the verbal

pain score was lowered one or more levels for at least two successive periods, either during the period of application (5 minutes) and at or after the 15 minute and 30 minute times.

At the completion of the experiment the appropriate treatment was rendered to alleviate the pain. No reports of allergy or adverse reaction to the applied medication were noted.

RESULTS

For the 27 experimental teeth the median verbal descriptor score at baseline was 2, the median cold score was 3, and the mean EPT score was 27.4 ± 3.31 (SEM). The values for paired control teeth were verbal = 0, cold = 1 and EPT = 32.3 ± 3.30 . After application of the medications, lower pain scores were reported by seven out of nine subjects treated with 10% benzocaine, seven out of nine with 20% benzocaine, and three out of nine with placebo. Because of the small sample size and since no difference was demonstrated between the active treatments, the results were combined from the two benzocaine groups, obtaining 14 of 18 subjects with pain relief after benzocaine. Compared to three of nine with pain relief following application of a placebo, the increased frequency of pain relief when an active medication was used was statistically significant ($p < 0.032$, Fisher exact test). On this basis, the hypothesis that benzocaine gels are better than placebo in causing pain relief, as judged by verbal descriptors, was accepted. This pilot study apparently had neither a large enough sample size nor a dependable method of measurement of local anesthetic duration that would allow distinction between 20% and 10% benzocaine gels, as was demonstrated previously.⁹ Four subhypotheses (SH) were also tested in this study:

SH-1: at baseline, teeth in pain should have lower response levels to EPT than control teeth. There were 15 experimental teeth with lower mean EPT scores than controls, one the same and 11 were higher than controls. Using a sign test, this set of outcomes is not significant, providing no substantial evidence that the results of EPT scores follow the directional hypothesis of SH-1.

To further investigate the sources of variability, an analysis of variance was run on the EPT scores at time 0. A large interaction term ($F = 29.20$, $p < 0.0001$) for patient by control tooth versus experimental tooth (i.e., how different patients react to the EPT test) was found; since the mean square of the interaction term (617.18) is larger than the mean square of the error term (21.14), the former should be used in testing for differences between mean EPT scores. Such a large mean square makes it difficult to detect a change in EPT after treatment with benzocaine. The analysis of variance results in a F ratio of 1.54, and

Table 1. Comparison of Control and Experimental EPT Scores After 15 Minutes

Pain Relief	$E > C$	$E < C$	$E = C$
Yes	12	4	1
No	0	10	0

failure to accept the hypothesis of a difference between experimental and control teeth in the same patient.

SH-2: Teeth in pain (experimental) should respond with higher discomfort levels to cold testing than control teeth. Pairing the experimental teeth with the control teeth, 25 of the 27 experimental teeth had higher cold scores at baseline than controls, one experimental was lower and one was the same. Using a sign test, this outcome has a probability of less than 0.0001 under the assumption that teeth in pain would respond with the same discomfort levels as control teeth as measured by cold testing, leading to acceptance of SH-2.

SH-3: Teeth relieved of pain should have higher EPT scores compared to control teeth. As an indication of direction of EPT scores, the average of 3 EPT scores taken 15 minutes after application of the test gel yielded the results in Table 1. The test of the hypothesis SH-3 concerns 17 patients who experienced pain relief. A sign test gives the probability of the rare occurrence in Table 1 as 0.038, indicating that after 15 minutes, patients who experienced pain relief also had a mean elevation of the EPT score, as compared to the paired control at the same time.

To further investigate the sources of variability, an analysis of variance was run with treatment gels as the factor of interest. This analysis shows that a large interaction term ($p < 0.0001$) causes a large variability in EPT response even from control to experimental from patient to patient (similar to that seen previously here). Our conclusion from this analysis is that EPT scores have too much inconsistency among patients for use in predicting pain relief, even though the overall direction of EPT change seems correct.

SH-4: Teeth that are relieved of pain should respond with lower discomfort levels to cold testing. Results from cold score testing are given in Table 2. Cold scores outcomes that were the same or increased were grouped into

Table 2. Incidence of Cold Score Changes by Pain Relief

Pain Relief	Cold Score	
	Lower	Same/Increased
Yes	12	5
No	4	6

one classification to avoid a larger contingency table with entries of 0 and 1 in two of the cells. Application of Fisher exact test to determine the independence of Cold Score and Pain Relief gives a probability of 0.223, failing to confirm SH-4.

DISCUSSION

Our results support the data of Sveen et al.⁸ and others⁹ that both benzocaine preparations are better than placebo for relieving toothache pain. In the present study there was a high percentage relief (78%) from either preparation, while the placebo effect was only 33%. By combining data from the two active treatments, which appeared similar in our study, a statistically significant increase in pain relief was obtained for the actives. The data support the use of benzocaine as an OTC toothache relief agent until the dentist can be seen for professional care. Our results do not disagree with previous findings that 20% benzocaine is more effective than 10% because our sample size was small and there was no direct measurement of duration of local anesthesia. Studies from several institutions with larger patient samples showed that the 20% gel was more efficacious than the 10% gel.⁹ This difference is difficult to demonstrate unless larger numbers of patients per medication are studied.

Thermal responses and EPT were not found to be helpful in predicting the patients pain severity or relief of discomfort. Among the three indicators used, verbal descriptors of pain appear to be the best predictors of clinical pain and of pain relief. The thermal cold scores, as used in this study, were highly correlated to the verbal descriptors in prediction of tooth status (pulpitis versus control) before treatment; however, cold scores were not useful in predicting pain relief because there was no statistically significant reductions of scores after treatment. Also,

EPT scores were not good predictors of either pain status or pain relief under the conditions of this study.

These studies should be continued from the standpoint of other in-office uses (for example root planning, topical anesthesia, etc.) and degree of pain relief during pulpectomy.

REFERENCES

1. Monheim's Local Anesthesia and Pain Control in Dental Practice. St. Louis, CV Mosby Co., 5th Edition, 1974, p 152.
2. Adriani J: Some aspects of the pharmacology of local anesthetics of clinical importance. *Marquette Med Rev* 1964; 30:46-52.
3. Adriani J, Zepernick R: Clinical effectiveness of drugs used for topical anesthesia. *JAMA* 1964;188:711-716.
4. Tainter M, Thronson AH, Moose SM: Studies in topical anesthesia II. Further observations on the efficacy of the more common local anesthetics when used on the gums and oral mucosa. *J Am Dent Assoc Cosmos* 1937;24:1480-1497.
5. Dalili H, Adriani J: The efficacy of local anesthetics in blocking the sensations of itch, burning, and pain in normal and "sunburned" skin. *Clin Pharmacol Ther* 1971;12:913-919.
6. North American Contact Dermatitis Group: Epidemiology of contact dermatitis in North America: 1972, *Arch Dermatol* 1973;108:537-540.
7. Gangarosa LP, Aleo JJ, Hurst V, Plein JB, Raymond PE, Scholle RH, Van Kirk LE: Relief or oral discomfort drug products for over-the-counter human use. OTC oral health care and discomfort drugs: establishment of a monograph. *Fed Reg (Proposed Rules)* 1982;47(101):22711-22930.
8. Sveen OB, Yaekel M, Adair SM: Efficacy of using benzocaine for temporary relief of toothache. *Oral Surg Oral Pathol Oral Med* 1982;53(6):574-576.
9. Orajel Studies, Data on File, Del Laboratories Inc., 565 Broad Hallow Road, Farmingdale, NY 11735.
10. Gangarosa LP: Electrical medication (iontophoresis): a modality for expanding dental practice with new therapies. *J Gen Dent* 1988;36:402-404.